



## Module 1

2 marks

1. What is IoT? Give real time examples.
2. Write any 2 Evolutionary Phases of the Internet.
3. List and brief Current Challenges Being Addressed by Connected Roadways.
4. List what are the main challenges facing manufacturing in a factory environment.
5. What are the Four Industrial Revolutions?
6. Write the functions of Applications layer.
7. Write the functions of Data abstraction layer.
8. List the seven-layers of IoT architectural reference model published by IoTWF.
9. Write the functions of Edge Computing Layer.
10. List different Access Technologies and Distances.
11. What are Point-to-point topologies.
12. What are Point-to-multipoint topologies.
13. Compare Star and Clustered Star Topology.
14. List the Sensor Types.
15. Define smart objects.

5 marks

1. Define digitization? Explain IoT and digitization.
2. Write a note on IoT impact.
3. Describe the framework for the Digital Ceiling.
4. Write a short note on Convergence of IT and OT.
5. Compare Operational Technology (OT) and Information Technology (IT).
6. List and explain IoT Challenges.
7. Explain the challenges in IoT Architectural Drivers.
8. Explain the Simplified IoT Architecture.
9. Explain the different layers of The Core IoT Functional Stack.
10. Write a note on fog computing.
11. Write a note on edge computing.
12. With suitable diagram explain How Sensors and Actuators Interact with the Physical World?
13. Give Actuator Classification by Energy Type with examples.
14. Write a note on Wireless Sensor Networks (WSNs).
15. List and describe Protocol Stacks Utilizing IEEE 802.15.4.
16. Draw the IEEE 802.15.4 MAC Format.

## Module 2

2 marks

1. List the Key Advantages of Internet Protocol.
2. Define Adaptation and Adoption.
3. Compare Bidirectional versus unidirectional data flow.
4. What is Fragmentation.
5. What is Transmission Control Protocol (TCP)?



6. What is User Datagram Protocol (UDP)?
7. What is the quality of service (QoS) that MQTT offers?
8. Draw MQTT Message Format.
9. Write a short note on IPV4.
10. Write a short note on IPV6.

5 marks

1. Explain why optimization is necessary for IP.
2. Write a note on Transport Layer.
3. Explain IoT Data Broker with suitable diagram.
4. Write a note on SCADA.
5. Explain Protocol Stack for Transporting Serial DNP3 SCADA over IP.
6. Explain High-Level IoT Protocol Stack for CoAP and MQTT.
7. Describe CoAP Message Format.
8. Write a short note on CoAP.
9. What is MQTT? Explain MQTT Publish/Subscribe Framework.
10. Tabulate MQTT Message Types.
11. Compare between CoAP and MQTT.

### MODULE 3

2 marks questions.

1. Compare Structured Versus Unstructured Data.
2. Compare Data in Motion Versus Data at Rest.
3. What is supervised learning.
4. What is unsupervised learning.
5. Define Integrity and Availability.
6. Write the diagram for Distributed Analytics Throughout the IoT System.
7. Write the diagram for Correlating Data Streams with Historical Data.
8. What is Edge Streaming Analytics?
9. How Apache Kafka Data Flow works.
10. Which are the NoSQL Databases.

5 marks questions.

1. What are the IoT Data Analytics Challenges. Explain.
2. Write a note on machine learning.
3. What is the importance of Neural Networks? Explain with example.
4. Explain “three Vs” of Big data.
5. Describe four major domains of ML used in IoT.
6. Write a note on Hadoop.
7. Explain Lambda Architecture.
8. List and explain Common Challenges in OT Security.
9. Define OCTIVE and explain OCTAVE Allegro Steps and Phases.
10. Write a note on FAIR.

## Module 4



2 marks question.

1. Which are the basic building blocks of IoT devices.
2. Why Arduino is used and where?
3. Describe the types of Arduino's.
4. How to install Arduino IDE.
5. What is the use of raspberry pi?
6. List variables and data types of Arduino.
7. Write a short note on DS18B20 Temperature Sensor.
8. How to Configure Raspberry Pi.
9. How to access Raspberry Pi remotely.
10. How to connect Raspberry Pi via SSH.

5 marks questions.

1. Write sample code to demonstrate Interfacing LED and switch with RaspberryPi.
2. Write a note on Arduino UNO.
3. Write a note Raspberry PI.
4. Give the pictorial representation for Raspberry pi mention all the GPIO pins.
5. What are features and applications of Raspberry PI.
6. Write a note on Raspberry pi generations.
7. What are features and applications of Arduino.
8. Give the sample Arduino code to switch on the LED.
9. How to implement Wireless Temperature Monitoring System Using Pi.
10. Describe Smart City IoT Architecture.
11. Write any 2 use cases for smart city.
12. Write a note on Smart City Security Architecture.